b) Using the construction in (a), find the function f^* corresponding to each partial function in Exercise 79.

Soln: 1. $f: \mathbb{Z} \to \mathbb{R}$, $f(n) = 1/n$.

$$f^*(a) = \begin{cases} f(a) = 1/a, \text{ where } a \in \mathbb{Z} - \{0\} \\ \sqrt{-1}, \text{ where } a = 0 \end{cases}$$ We Know, $\sqrt{-1} \notin \mathbb{R}$.

2. $f: \mathbb{Z} \to \mathbb{Z}$, $f(n) = \lceil n/2 \rceil$

$$f^*(a) = \{ \lceil a/2 \rceil, \text{ where } a \in \mathbb{Z} \}$$

3. $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Q}$, $f(m,n) = m/n$.

$$f^*(a,b) = \begin{cases} a/b, \text{ where } a \in \mathbb{Z} \text{ and } b \in \mathbb{Z} - \{0\} \\ \sqrt{2}, \text{ where } a \in \mathbb{Z} \text{ and } b = 0 \end{cases}$$ We Know $\sqrt{2} \notin \mathbb{Q}$

4. $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(m,n) = mn$

$$f^*(a,b) = \{ ab \text{ where } a \in \mathbb{Z} \text{ and } b \in \mathbb{Z} \}$$

5. $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(m,n) = m - n$ if $m > n$.

$$f^*(a,b) = \begin{cases} a - b, \text{ where } a \in \mathbb{Z} \text{ and } b \in \mathbb{Z}, a > b \\ 0.1, \text{ where } a, b \in \mathbb{Z}, a \not> b, \end{cases}$$ We Know $0.1 \notin \mathbb{Z}$.

81) a) Show that if a set S has cardinality m, where m is a positive integer, then there is a one-to-one correspondence between S and the set $\{1, 2, \ldots, m\}$

Soln: WLOG, Let S be $\{a_1, a_2, \ldots, a_m\}$.

Let $f: \{1, 2, \ldots, m\} \to \{a_1, a_2, \ldots, a_m\}$ and $f(i) = a_i$.

, where $i \in \{1, 2, \ldots, m\}$. It's easy to see that $\forall x \in \{a_1, a_2, \ldots, a_m\}$.

$\exists i \in \{1, 2, \ldots, m\}$ s.t. $f(i) = x$. By Defn. 7, f is onto.

By Prob 74, f is one-to-one. $\therefore$ by Defn. 8, f is a one-to-one correspondence. $\square$